COMPUTER PROGRAMMING 3(Lec)

DATA QUERY

FROM THE DATABASE

MS ACCESS — Customer_Table

Field	Data Type	Key
Customer_ID	AutoNumber	
First_Name	Short Text	
Last_Name	Short Text	
City	Short Text	
Phone	Short Text	

GROUP 5 | BSIT-S-T-1B-T

NUÑEZ, CEDRIC HANZ

COSTELO, CHRISTIAN

PATATAG, ALFRED

ODOÑO, DHAN ANGELO

What is a Query?

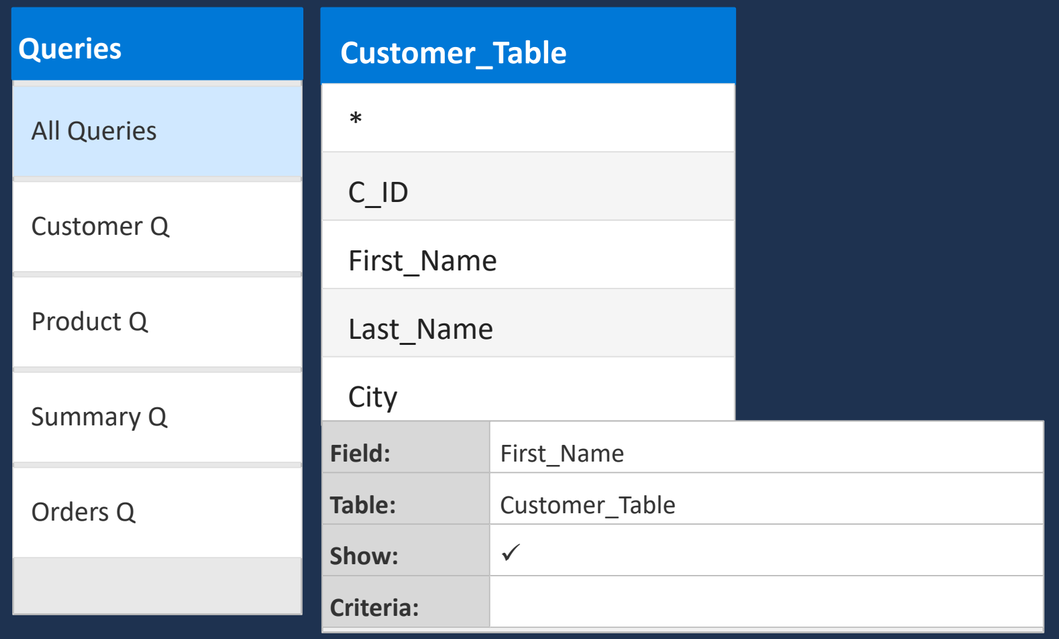
In databases, a query may retrieve information from multiple tables and combine it for output in a form or report. A query can either obtain data results from a database, execute action on the data, or both.

Queries are able to fulfill simple inquiries, do computations, aggregate data from many tables, and add, edit, or delete data. Because queries are so flexible, there are numerous types — and you may create one depending on the task.

A query can:

- Retrieve data from single or multiple tables
- Execute actions on stored data
- Perform calculations and aggregations
- Add, edit, or delete records

Query in MS Access



Result Datasheet:

First_Name	Last_Name	City
Jenny	Stones	Taguig
Bill	Brown	Taguig
Daisy	Campbell	Taguig

Types of Queries

Parameter Query

User enters criteria into a prompt box to filter records. Ideal for reusing the same query with different conditions each time.

Multitable Query

Combines data from two or more connected tables using common key fields. Also called joining tables.

Summary Query

Aggregates data using functions such as SUM, AVG, MIN, MAX, and COUNT. Essential for reports and totals.

Crosstab Query

Summarizes data in a compact pivot-table format with both row and column headings.

Action Query

Modifies database data — four types: Make Table, Append, Update, and Delete.

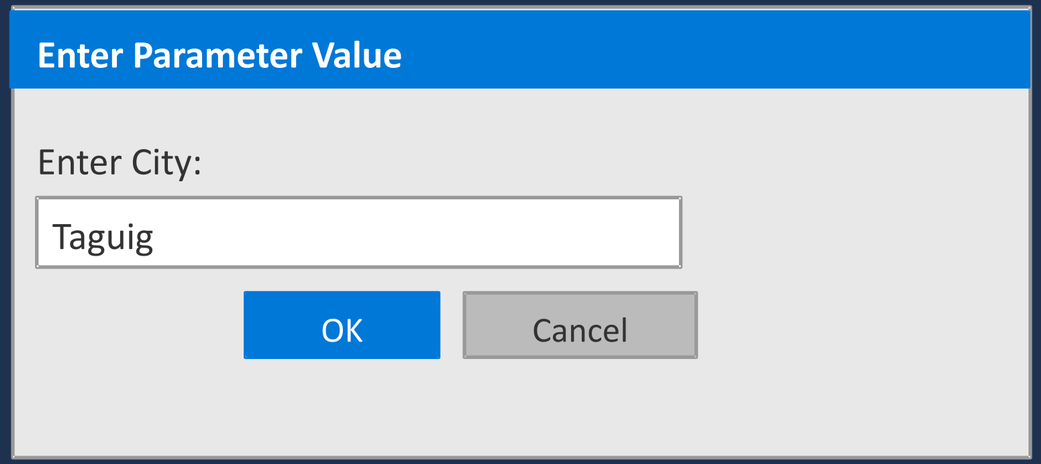
Parameter Query

A parameter query is one in which the user enters parameters or criteria into an input box to choose a collection of records.

A Parameter Query is convenient for repeating the same query but with different conditions. In simple terms, it means executing the exact same set of queries with various requirements.

- Flexible and reusable query design
- Prompts user for input at runtime
- Returns different results from the same template
- Reduces need to create multiple separate queries

How It Works



Query Design — Criteria row:

	First_Name	Last_Name	City
Field:	First_Name	Last_Name	City
Table:	Customer_Table	Customer_Table	Customer_Table
Show:	✓	✓	✓
Criteria:			[Enter City]

Result (City = Taguig):

First_Name	Last_Name	City
Jenny	Stones	Taguig
Bill	Brown	Taguig

Creating a Query — Overview

In Microsoft Access, a query is created using the Query Design view. This tool allows you to visually select tables, choose fields, set sort orders, and specify criteria — without writing any code.

01

Open Query Design

Go to the Create tab on the Ribbon → click Query Design in the Queries group. This opens the Query Design view.

02

Select Table(s)

The Show Table dialog opens. Select the table(s) you need, click Add, then click Close to proceed.

03

Add Fields & Run

Double-click or drag fields to the Design Grid. Click Run (!) on the Design tab to display results in Datasheet View.

💡 Tip: You can also use the Query Wizard (Create tab) for a step-by-step guided approach to building queries.

Query Design View

The Query Design view is divided into two areas: the Table Area (top) showing table field lists, and the Design Grid (bottom) where you define the query structure.

Query Design View

Customer_Table
C_ID
First_Name
Last_Name
City
Phone

Table Area (field lists)

	First_Name	Last_Name	City
Field:	First_Name	Last_Name	City
Table:	Customer_Table	Customer_Table	Customer_Table
Sort:			
Show:	✓	✓	✓
Criteria:			

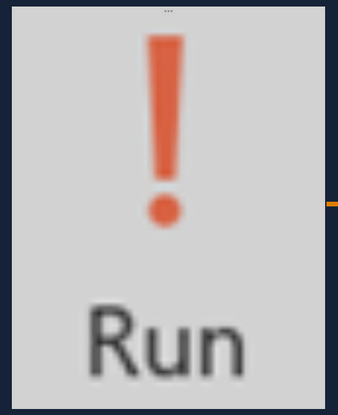
Design Grid (define query)

Table Area

Displays table boxes you added. Each box lists all available fields you can add to the query.

Design Grid Rows

Field, Table, Sort, Show, Criteria, Or — each row controls a different aspect of the query.



Run Button (to be click)

Running the Query

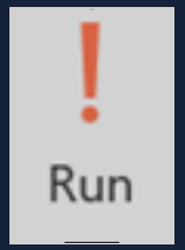
Click the Run (!) button on the Design tab to execute and view results in Datasheet View.

Query Results & Datasheet View

After running a query, results are displayed in Datasheet View — a spreadsheet-like grid showing only the fields and records that matched the query criteria.

Design Grid (query definition):

	First_Name	Last_Name	City
Field:	First_Name	Last_Name	City
Table:	Customer_Table	Customer_Table	Customer_Table
Sort:			
Show:	✓	✓	✓
Criteria:			



Datasheet View (results):

First_Name	Last_Name	City
Jenny	Stones	Taguig
Simon	Cowell	Manila
Haru	Stones	Manila
Lenard	Moore	Taguig
Matthew	Lopez	Pasay
Mike	Johnson	Quezon
Ann	Smith	Manila
Bill	Brown	Manila

💡 With queries, data retrieval becomes easier — whether from a single table or multiple joined tables.

Modifying a Query — Sorting

You can modify an existing query design to control the order of results. In the Sort row of the Design Grid, select Ascending or Descending for the desired field. Multiple fields can be sorted simultaneously — the leftmost sorted field takes priority.

Design Grid — Sort applied:

	C_ID	First_Name	Last_Name
Field:	C_ID	First_Name	Last_Name
Table:	Customer_Ta	Customer_Table	Customer_Table
Sort:		Ascending ▼	
Show:	✓	✓	✓

Sorted Result (A→Z by First_Name):

C_ID	First_Name	Last_Name
C14	Adrian	Hudson
C07	Ann	Smith
C08	Bill	Brown
C12	Daisy	Campbell
C03	Haru	Stones
C10	John	Young
C09	Laura	Paterson

Modifying a Query — Criteria

AND Criteria — same row, both conditions must be true

Place all conditions on the same Criteria row. Only records matching ALL criteria are returned.

	First_Name	Gender	Age
Field:	First_Name	Gender	Age
Criteria:		"Male"	> "25"

First_Name	Gender	Age
Neil	Male	40
Haru	Male	30
Mike	Male	35
Bill	Male	27

OR Criteria — different rows, either condition can be true

Place conditions on separate rows. Records matching ANY of the criteria rows are returned.

	Gender	Age
Criteria:	"Male"	
or:		>25

First_Name	Gender	Age
Neil	Male	40
Haru	Male	30
Mike	Male	35
Bill	Male	27
Sarah	Female	29
Emily	Female	26

Criteria Functions Reference

Criteria Function	Format Syntax
Is x	`"x"`
Is not x	`Not In ("x","y")`
Begins With x	`Like "x*"`
Ends With x	`Like "*x"`
Contains x	`Like "*x*"`
Does Not Contain x	`Not Like "*x*"`

Criteria Function	Format Syntax
Between x and y	`Between x And y`
Equals	`x`
Does Not Equal	`<> x`
Greater Than / ≥	`> x  or  >= x`
Less Than / ≤	`< x  or  <= x`
Input Parameter	`[Enter Display Text]`

💡 AND = same Criteria row | OR = different rows | Input Parameter = [Text in brackets]

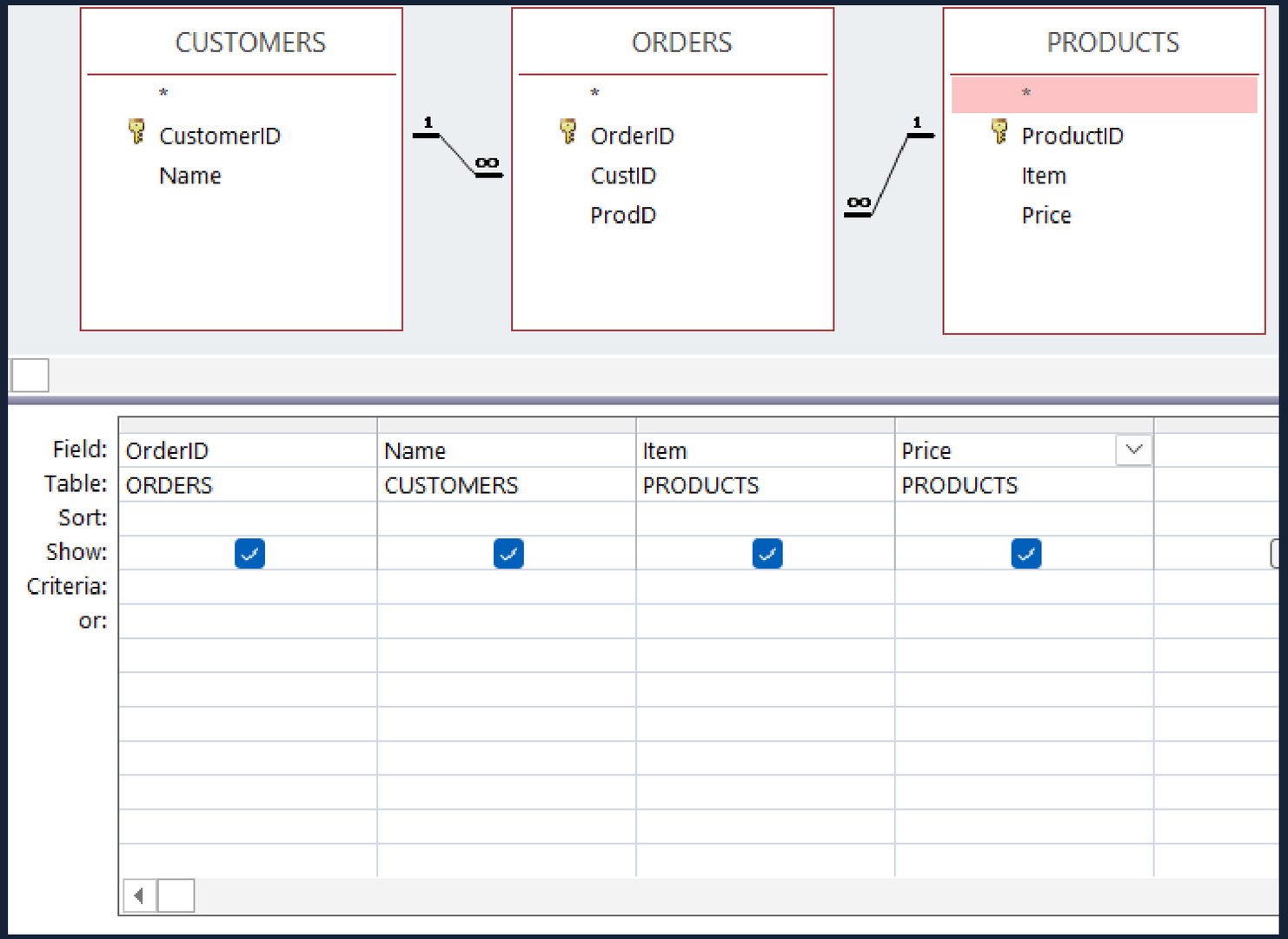
Multitable Query - Definition

A multi-table query in Access is a query that retrieves data from two or more tables simultaneously. This capability is essential for relational databases, where data is often distributed across multiple tables to reduce redundancy and improve data integrity. By using multi-table queries, users can combine related data from different tables into a single result set, making it easier to analyze and report on the information.

Multitable Query - Example

Tables and their fields:

- CUSTOMERS- CustomerID (PK), Name
- ORDERS- OrderID (PK), CustID, ProdID (foreign keys)
- PRODUCTS- ProductID (PK), Item, Price



Multitable Query - Datasheet Result

OrderID	Name	Item	Price
O01	Alperd	Sofa	₱8,000.00
O02	Shon	Electricfan	₱1,250.00
O03	Luwins	Aircon	₱12,000.00
O04	Betia	Lamp	₱600.00
*			

Joining Tables - Definition

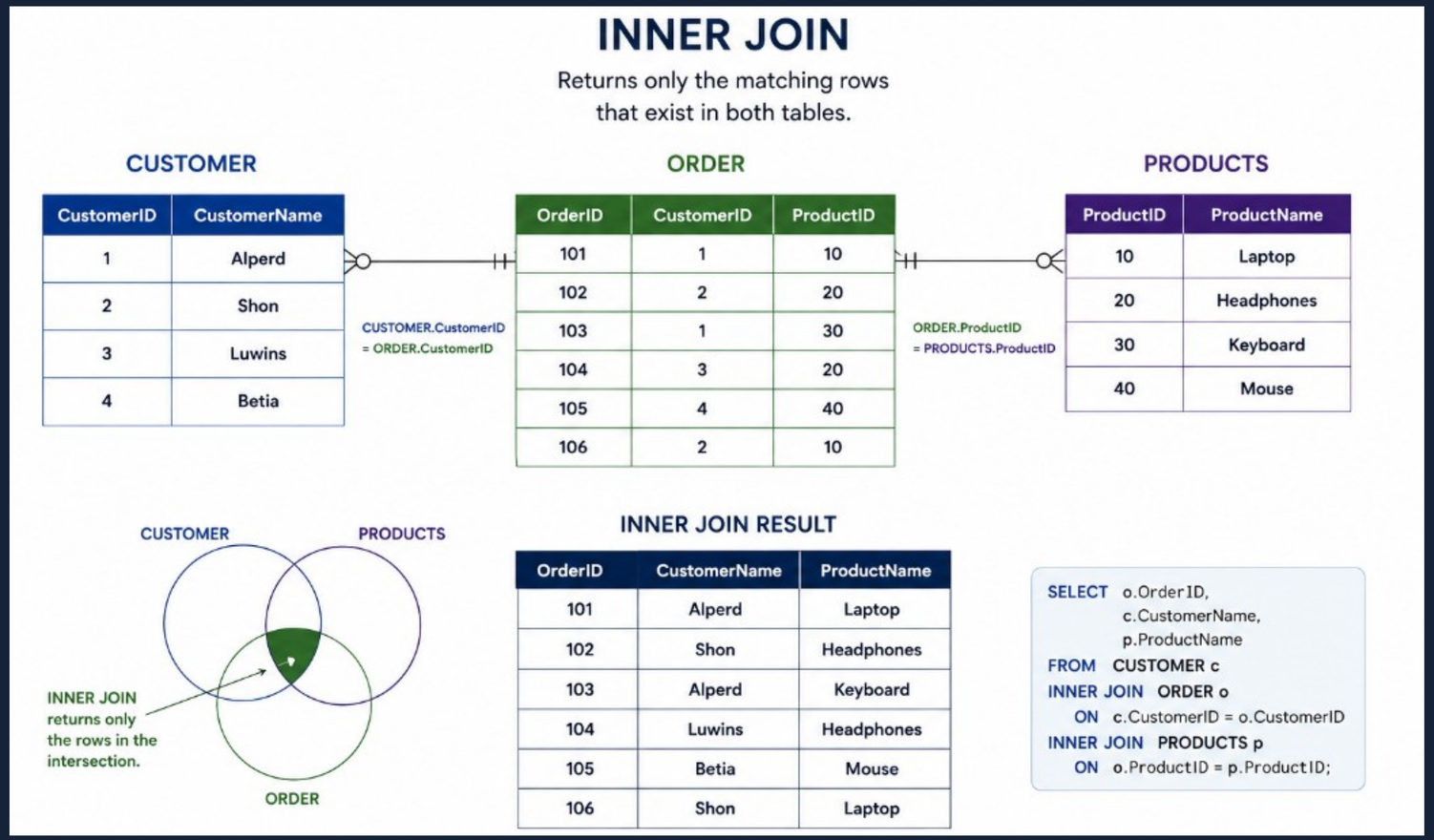
Joining tables is the process of linking two or more tables together using a common field, such as a primary key and a foreign key, to combine their data into one result.

Types of join

1. Inner Join
2. Left Outer Join
3. Right Outer Join
4. Full Outer Join

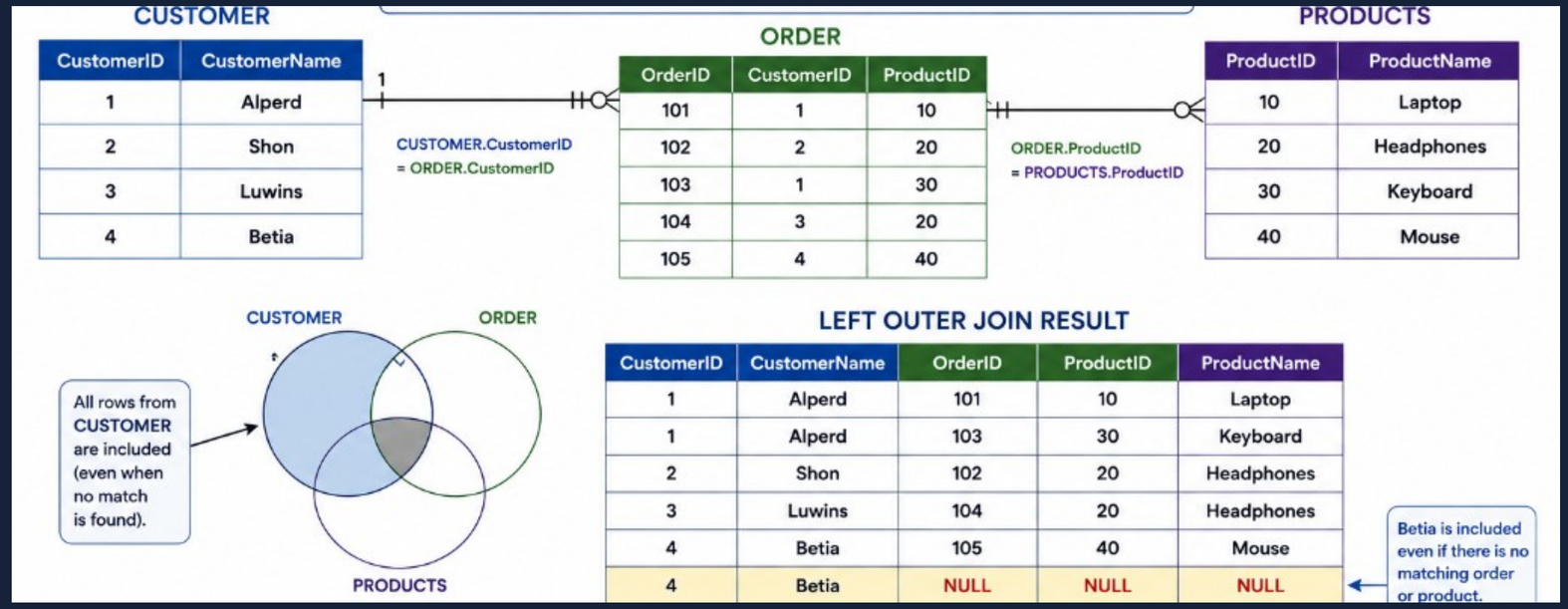
1. Inner Join

An Inner Join is the type of join that returns only the records that have matching values in both tables.



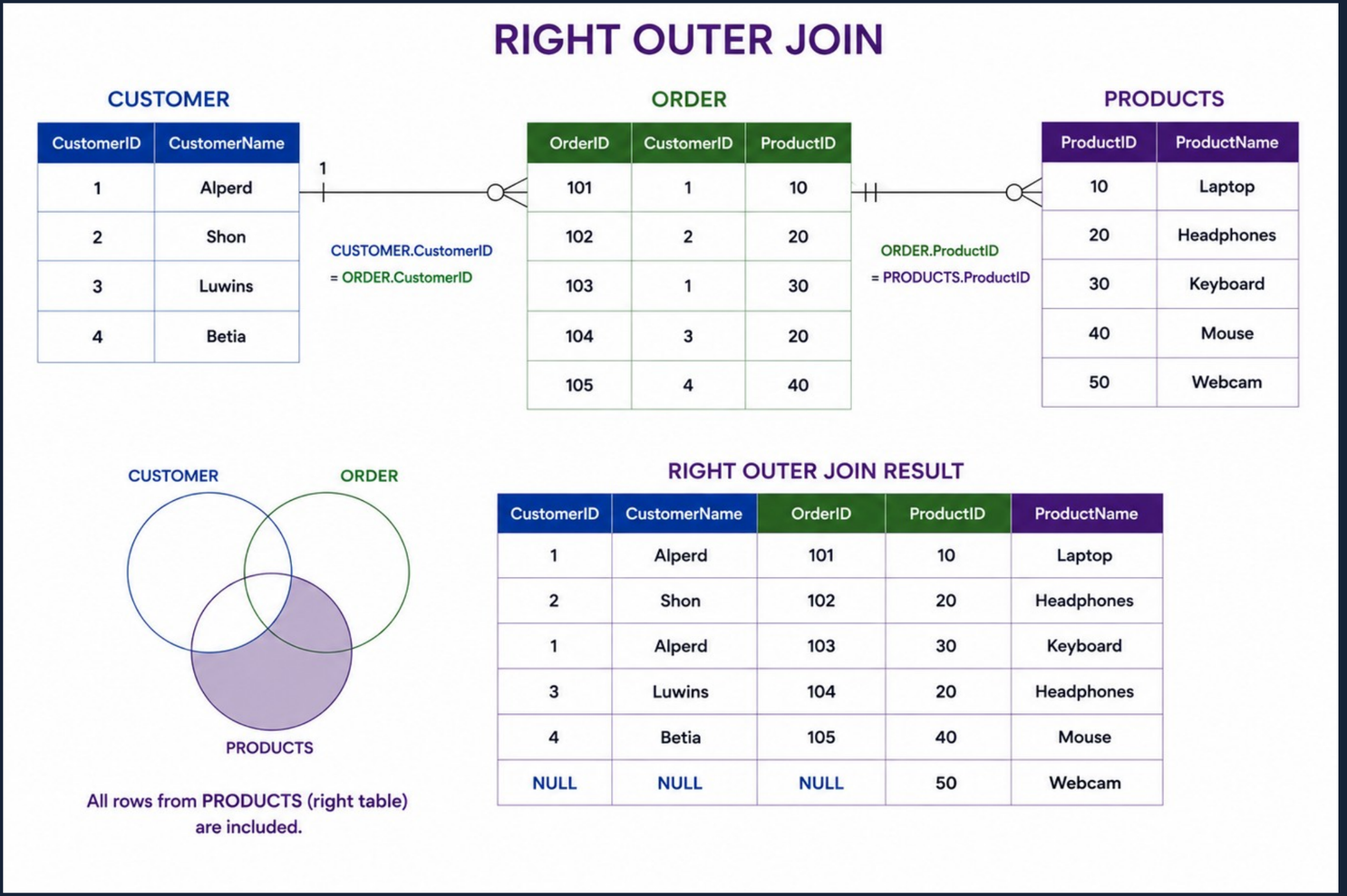
1. Left Outer Join

A Left Outer Join returns all records from the left table and only the matching records from the right table.



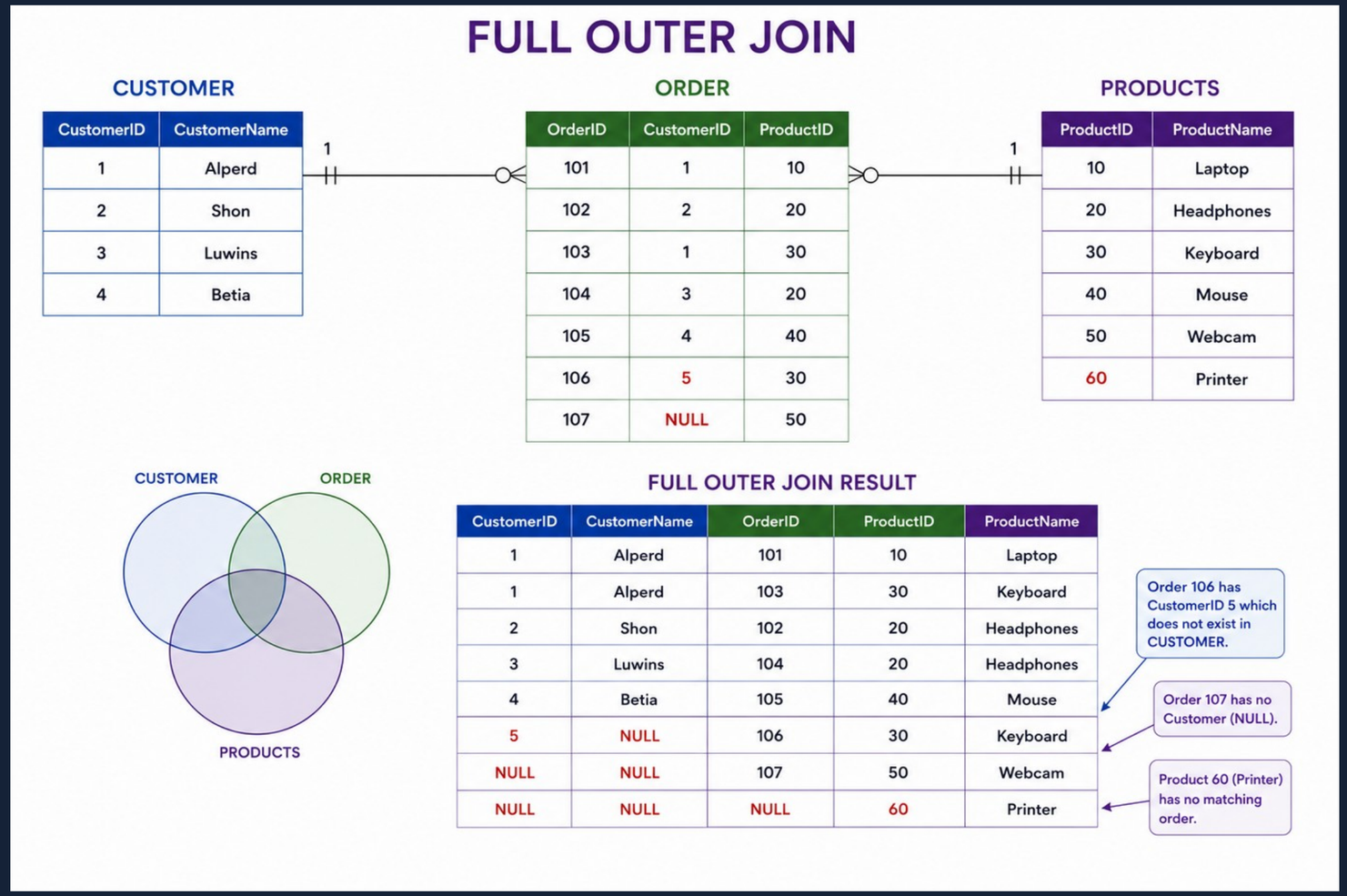
1. Right Outer Join

A Right Outer Join returns all records from the right table and only the matching records from the left table.



1. Full Outer Join

A Full Outer Join returns all records from both the left table and the right table.



Summary Query - Definition

Summary query is a database query used to calculate and display summarized information from multiple records.

Summary queries can be used in:

- **Reports and dashboards-** To show totals, averages, counts, or other summaries (ex. total sales per month).
- **Business analysis-** To understand performance like total revenue, number of customers, or average order value.
- **Grouping data-** When you want results grouped by a category (ex. sales per product, students per course).
- **Data analysis-** Example: total grades per student, number of patients per doctor, etc.

Summary Query - Functions

- **SUM–** Adds all values in a field.
- **COUNT–** Counts the number of records in a field.
- **AVG–** Calculates the average value of a field.
- **MIN–** Returns the smallest value in a field.
- **MAX–** Returns the largest value in a field.

Crosstab - Definition

A **Crosstab Query** is used to summarize and analyze information by arranging data into a grid format with rows and columns. It applies aggregate functions such as sum, average, or count to produce meaningful summaries.

The main purpose of a crosstab query is to make large datasets easier to interpret by presenting results in a matrix, allowing users to quickly compare values across different categories and time periods.

Crosstab - Structure

01

Open the Crosstab Query Wizard

Start the process from the Create tab in MS Access.

Go to

Create → Query Wizard → Crosstab Query Wizard- Select Crosstab Query Wizard from the list

- Click OK to begin

02

Choose the Source Table

Select the table or query that contains the data you want to analyze.

- For example, choose the Sales Table
- This table links to Customers and Products through IDs

03

Select Row Headings

Pick the fields that will appear as rows in the matrix.

04

Select Column Headings

Pick the field that will appear as columns across the top.

- Choose one field only
- Example: Location from the Customers Table

05

Select Values to Summarize

Choose the numeric field and the aggregate function to calculate.

- Example: TotalAmount from the Sales Table
- Functions: Sum, Avg, Count

06

Name and Finish

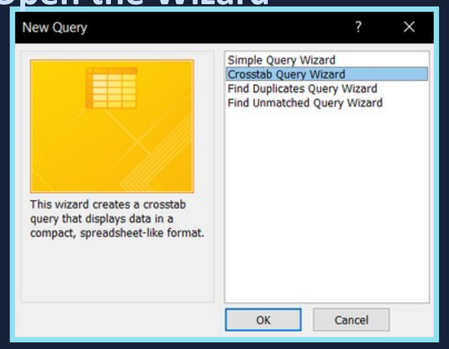
Save and run the query to view results.

- Enter a name (e.g., Product Sales by Location)
- Click Finish to display the crosstab matrix

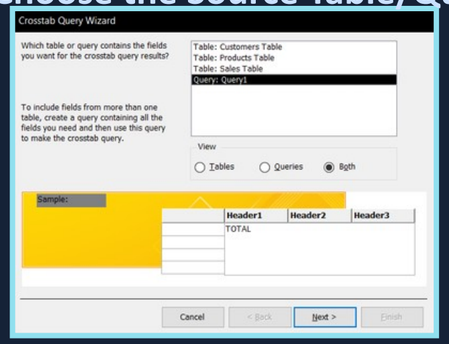
Crosstab - Example: Sales by Product and Location

Steps:

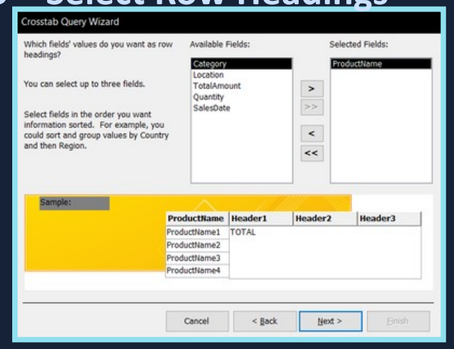
01 – Open the Wizard



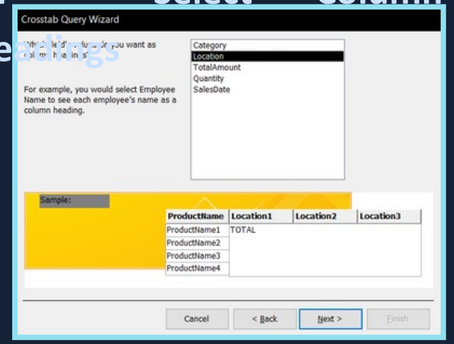
02 – Choose the Source Table/Query



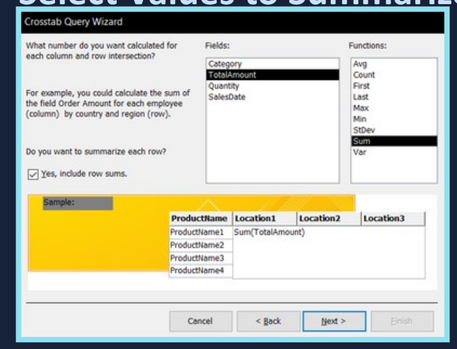
03 – Select Row Headings



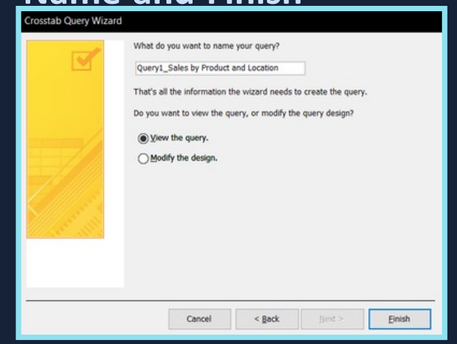
04 – Select Column Headings



05 – Select Values to Summarize



06 – Name and Finish

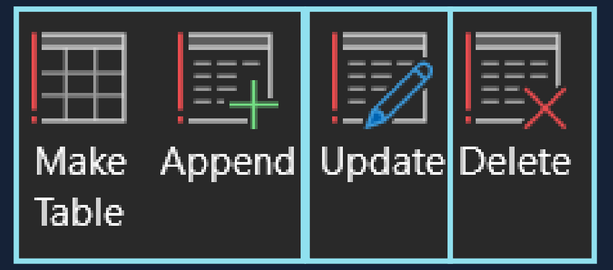


Result:

Query1 | Query1_Sales by Product and Location

ProductNam	Total Of Tota	Las Pinas	Paranaque
Soap	₱100.00	₱100.00	
Toothbrush	₱100.00		₱100.00

Action Queries - Definition



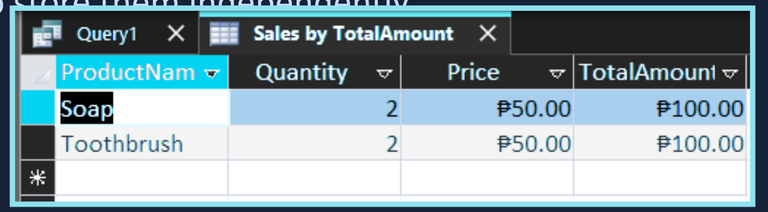
An Action Query in Microsoft Access is a type of query that directly modifies data in a database. Unlike select queries that only display results, action queries perform operations such as adding, updating, deleting, or creating tables. Their purpose is to automate bulk changes and make data management more efficient, but because they permanently alter records, they must be used carefully.

There are **four main types**: a **Make-Table Query** creates a new table from existing data, an **Append Query** adds records from one table or query into another, an **Update Query** changes values in existing records, and a **Delete Query** removes records based on specified criteria. For example, an update query can change all “Television” entries to “TV” in the Orders table, while a delete query can remove all records where the product name equals “Television.”

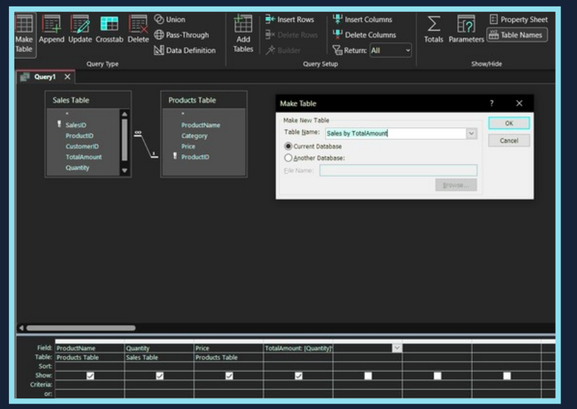
Make-Table Query

A Make-Table Query creates a new table from the results of a query. It is useful when you want to archive data, create backup copies, or generate a separate dataset for reporting and analysis.

For example, you can make a table of the sales by the total amount to store them independently.


Query1 | Sales by TotalAmount

ProductNam	Quantity	Price	TotalAmount
Soap	2	₱50.00	₱100.00
Toothbrush	2	₱50.00	₱100.00



Action Queries - Append Query

An Update Query modifies values in existing records based on specified criteria. It is often used for bulk changes, such as adjusting prices, correcting names, or updating categories.

For example, you can update the Products table to add other Items.

Before Query:

Destination Table:

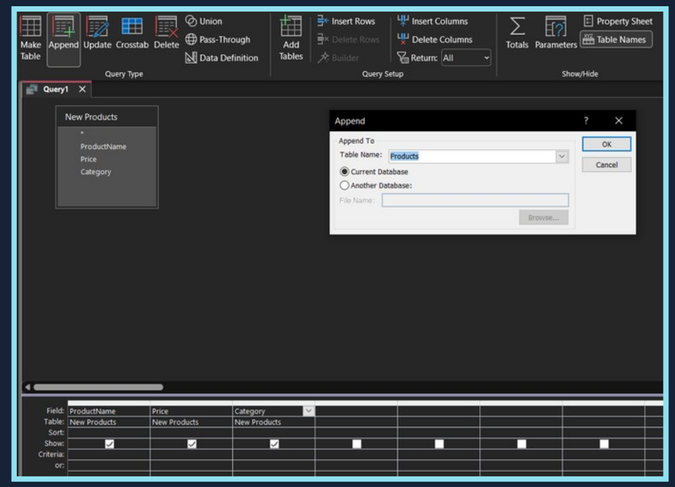
Products

ProductID	ProductNan	Price	Category	Click to Add
1	Kape	₱50.00	Beverage	
2	Pandesal	₱5.00	Food	
3	Cheese Pande	₱10.00	Food	
4	Pancit Canton	₱45.00	Food	
5	Milk Tea	₱50.00	Beverage	
(New)		₱0.00	Uncategorized	

Appending Table:

Products | New Products

ProductNan	Price	Category
Spanish Latte	₱75.00	
Mocha Latte	₱50.00	
Choco Mousse	₱55.00	
Matcha	₱45.00	



After Query:

Query1 | Products

ProductID	ProductNan	Price	Category
1	Kape	₱50.00	Beverage
2	Pandesal	₱5.00	Food
3	Cheese Pande	₱10.00	Food
4	Pancit Canton	₱45.00	Food
5	Milk Tea	₱50.00	Beverage
14	Spanish Latte	₱75.00	
15	Mocha Latte	₱50.00	
16	Choco Mousse	₱55.00	
17	Matcha	₱45.00	
(New)		₱0.00	Uncategorized

Action Queries - Update Query

An Update Query modifies values in existing records based on specified criteria. It is often used for bulk changes, such as adjusting prices, correcting names, or updating categories.

For example, you can update all product prices by 10% in the Products table.

Before Query:

Original Products Table:

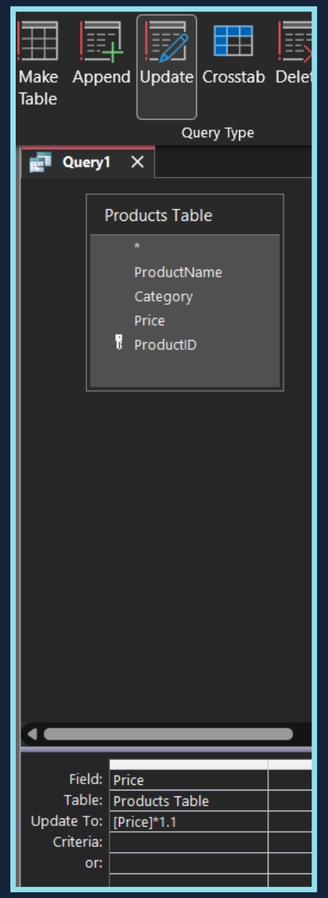
Products Table

ProductNam	Category	Price	ProductID
Soap	Hygiene	₱50.00	1
Toothbrush	Hygiene	₱50.00	2
Toothpaste	Hygiene	₱100.00	3
Mouthwash	Hygiene	₱100.00	4
Shoes	Clothing	₱2,000.00	5
Fries	Food	₱100.00	6
C2	Beverage	₱40.00	7
Bread	Food	₱100.00	8
T shirt	Clothing	₱1,500.00	9
Shampoo	Hygiene	₱70.00	10

After Query:

Query1 | Products Table

ProductNam	Category	Price	ProductID
Soap	Hygiene	₱55.00	1
Toothbrush	Hygiene	₱55.00	2
Toothpaste	Hygiene	₱110.00	3
Mouthwash	Hygiene	₱110.00	4
Shoes	Clothing	₱2,200.00	5
Fries	Food	₱110.00	6
C2	Beverage	₱44.00	7
Bread	Food	₱110.00	8
T shirt	Clothing	₱1,650.00	9
Shampoo	Hygiene	₱77.00	10
	Uncategorized		(New)



Action Queries - Delete Query

A Delete Query removes records that meet certain conditions. It is used to clean up or restructure data by eliminating unwanted records.

For example, you can delete all sales records where the quantity is zero, or remove customers who no longer have transactions.

Before Query:

Original Sales Table:

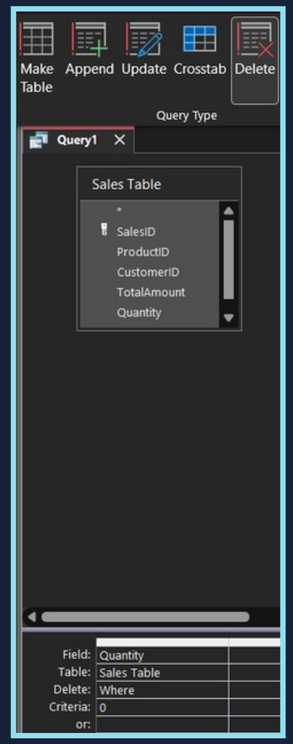
Sales Table

SalesID	ProductID	CustomerID	TotalAmour	Quantity	SalesDate
1	1	1	₱2.00	2	07/05/2026
2	2	2	₱1.00	2	07/05/2026
3	6	2	₱0.00	0	11/05/2026
(New)	0	0	₱0.00	0	11/05/2026

After Query:

Sales Table

SalesID	ProductID	CustomerID	TotalAmour	Quantity	SalesDate
1	1	1	₱2.00	2	07/05/2026
2	2	2	₱1.00	2	07/05/2026
#Deleted	#Deleted	#Deleted	#Deleted	#Deleted	#Deleted
(New)	0	0	₱0.00	0	11/05/2026



Using Multitable Queries - Definiton

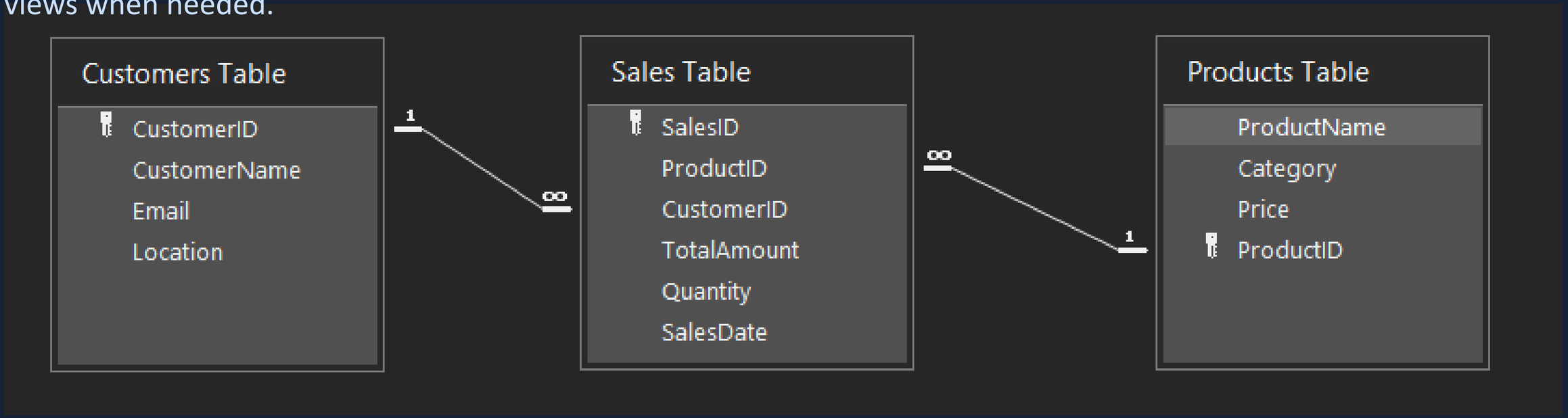
A Multitable Query is a query that combines data from two or more related tables in Microsoft Access. Instead of working with a single table, it uses relationships between tables to pull together information stored in different places.

This allows you to view and analyze data more effectively, especially when details are spread across multiple tables.

Using Multitable Queries - Structure

The structure of a multitable query relies on joins between tables, usually through a common field such as CustomerID or ProductID. These joins ensure that related records are matched correctly. The purpose of multitable queries is to provide richer, more meaningful results by combining attributes from different tables. They are especially useful for reporting, analysis, and building forms or reports that require data from multiple sources.

In practice, they help reduce redundancy by keeping data normalized across tables while still allowing comprehensive views when needed.



Using Multitable Queries - Example (TotalSpent)

Before Query:

Products Table

Products Table | Sales Table | Customers Table

ProductNam	Category	Price	ProductID	Click to Add
Soap	Hygiene	₱55.00	1	
Toothbrush	Hygiene	₱55.00	2	
Toothpaste	Hygiene	₱110.00	3	
Mouthwash	Hygiene	₱110.00	4	
Shoes	Clothing	₱2,200.00	5	
Fries	Food	₱110.00	6	
C2	Beverage	₱44.00	7	
Bread	Food	₱110.00	8	
T shirt	Clothing	₱1,650.00	9	
Shampoo	Hygiene	₱77.00	10	
	Uncategorized		(New)	

Customers Table

Customers Table | Sales Table | Products Table

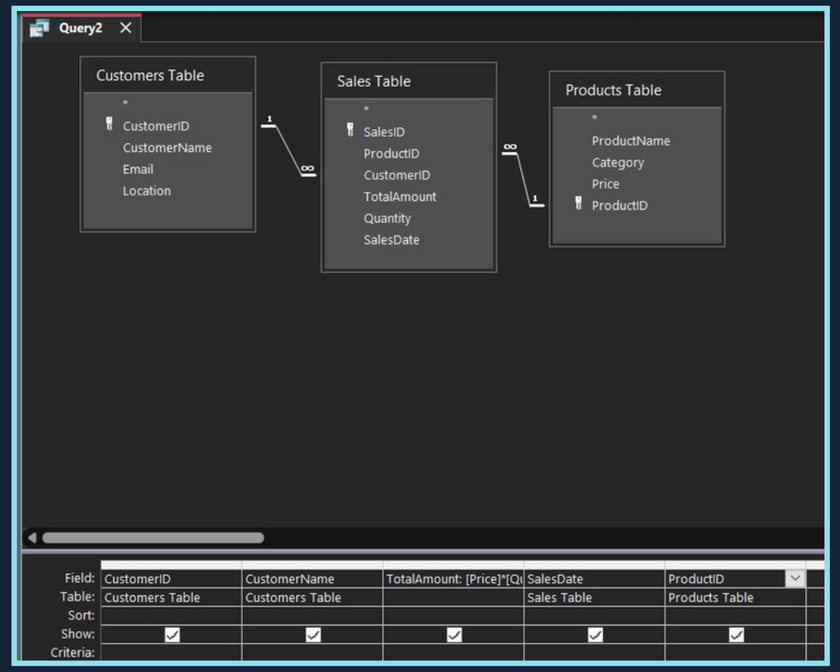
CustomerID	CustomerNa	Email	Location
1	Timothy Bapilar	Timothybapilar(	Las Pinas
2	Beitia Alfred	BeitiaA@gmail.	Paranaque
3	Michael Ballebe	Ballebears@gm	Pasay
4	Franklin Macara	FrankMg@gmai	Manila
5	Trevor Agustin	Tagus@gmail.c	Marikina
6	John John Abe	Swabedaddy@g	Pasig
7	James Lebron	TheGoat@gmai	Taga-Samin
(New)			

Sales Table

Customers Table | Sales Table | Products Table

SalesID	ProductID	CustomerID	TotalAmour	Quantity	SalesDate
1	1	1	₱2.00	2	07/05/2026
2	2	2	₱1.00	2	07/05/2026
(New)	0	0	₱0.00	0	11/05/2026

After Query:



Customers Table | Sales Table | Products Table | Query2

CustomerID	CustomerNa	TotalSpent	SalesDate	ProductID
1	Timothy Bapilar	₱110.00	07/05/2026	1
2	Beitia Alfred	₱110.00	07/05/2026	2
(New)				(New)

THANK YOU

Data Query from the Database

GROUP 5 | BSIT-S-T-1B-T | Computer Programming 3

Topics Covered

- 5.1 Types of Queries
- 5.2 Creating a Query
- 5.3 Modifying Design
- 5.4 Multitable Query
- 5.5 Joining Tables
- 5.6 Summary Queries
- 5.7 Crosstab Queries
- 5.8 Action Queries
- 5.9 Using Multitable